1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

ANS:-

Protocol names seen in Wireshark capture

- After installing Wireshark and capturing traffic while visiting instagram.com, the following protocol names appeared in the packet list (among others):
- Protocol Names (examples): DNS, IPv4, SSDP, TCP, IPv6, MDNS, TLS, ARP, DHCP, HTTP, ICMP, IGMP, QUIC, UDP, LLMNR
- *(Your capture may show slightly different protocols depending on your device and network.)*

2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

ANS:-

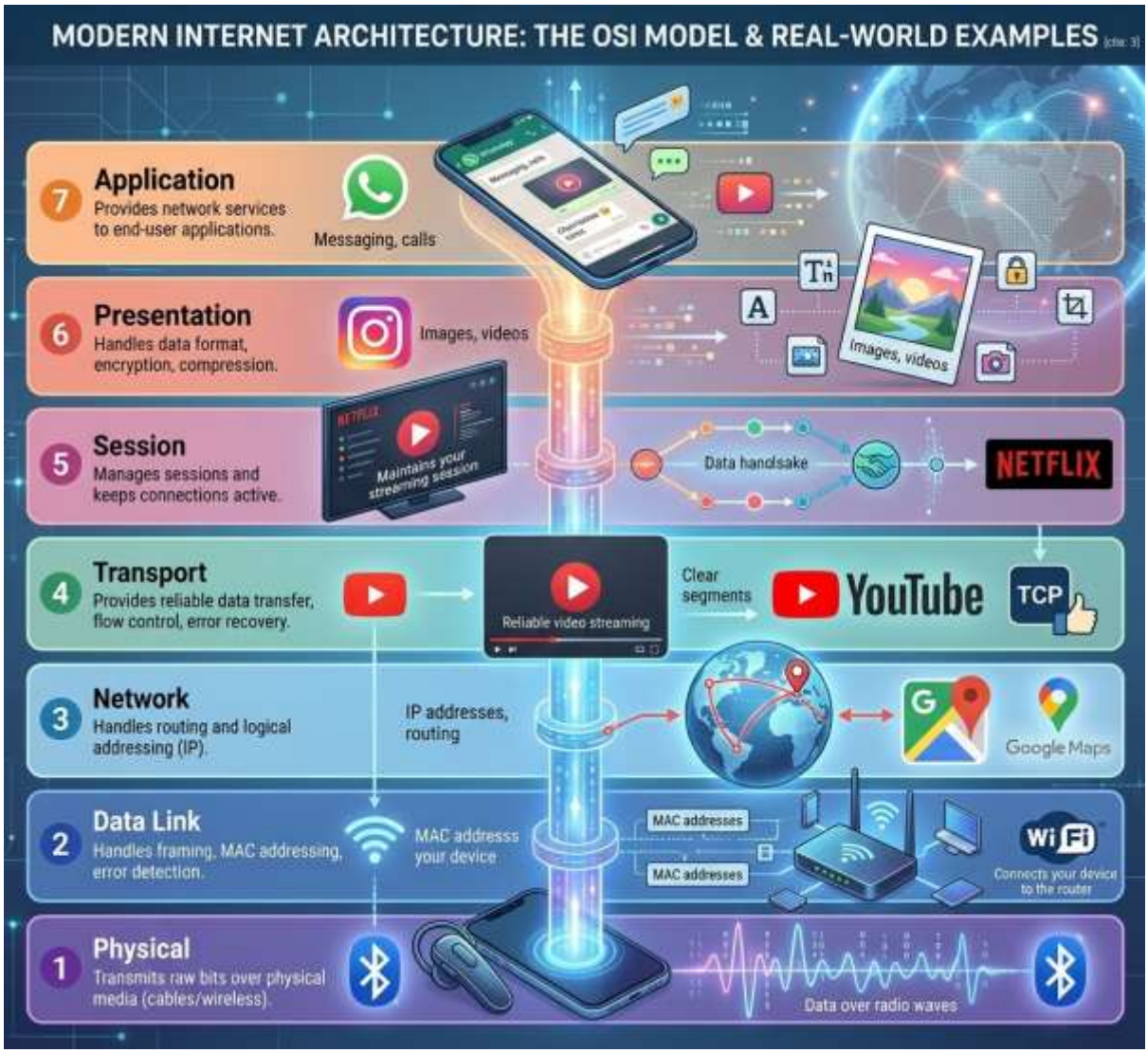
three packets and their OSI layers

Here are 3 packets from the capture and the OSI layer they mainly represent:

No.	Protocol	Info (example)	Main OSI Layer
1	HTTP	GET / HTTP/1.1	Application Layer (Layer 7)
2	TCP	443 → 52932 [ACK]	Transport Layer (Layer 4)
3	TLS	Client Hello	Presentation Layer (Layer 6)

3. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

ANS:-



**4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.**

ANS:-

When you send a message on WhatsApp, it goes through multiple layers of the OSI model, from your device to the recipient's device.

- **1. Application Layer (Layer 7):** You type the message in WhatsApp. The app prepares the message and uses WhatsApp's servers to deliver it. *(WhatsApp app)*
- **2. Presentation Layer (Layer 6):** The message is formatted (e.g., text encoding like UTF-8). If needed, it can be encrypted for secure communication (end-to-end encryption). *(Encryption (secure data))*

- **3. Transport Layer (Layer 4):** The data is split into segments. TCP ensures the data is delivered reliably and in order. *(TCP Reliable delivery)*
- **4. Network Layer (Layer 3):** The segments are wrapped in IP packets. IP addresses are used to route the data through the internet. *(IP routing)*
- **5. Data Link Layer (Layer 2):** The packets are framed and sent over your Wi-Fi or mobile network. MAC addresses help deliver the data to the next device (router). *(Wi-Fi / Mobile Network)*
- **6. Physical Layer (Layer 1):** The data is converted into electrical or radio signals. These signals travel through the air (Wi-Fi/cellular) to the internet. *(Radio signals (over the air))*

The recipient's device then receives the data, and the process happens in reverse (from Layer 1 to Layer 7) to display your message in WhatsApp.